PolygonRotation

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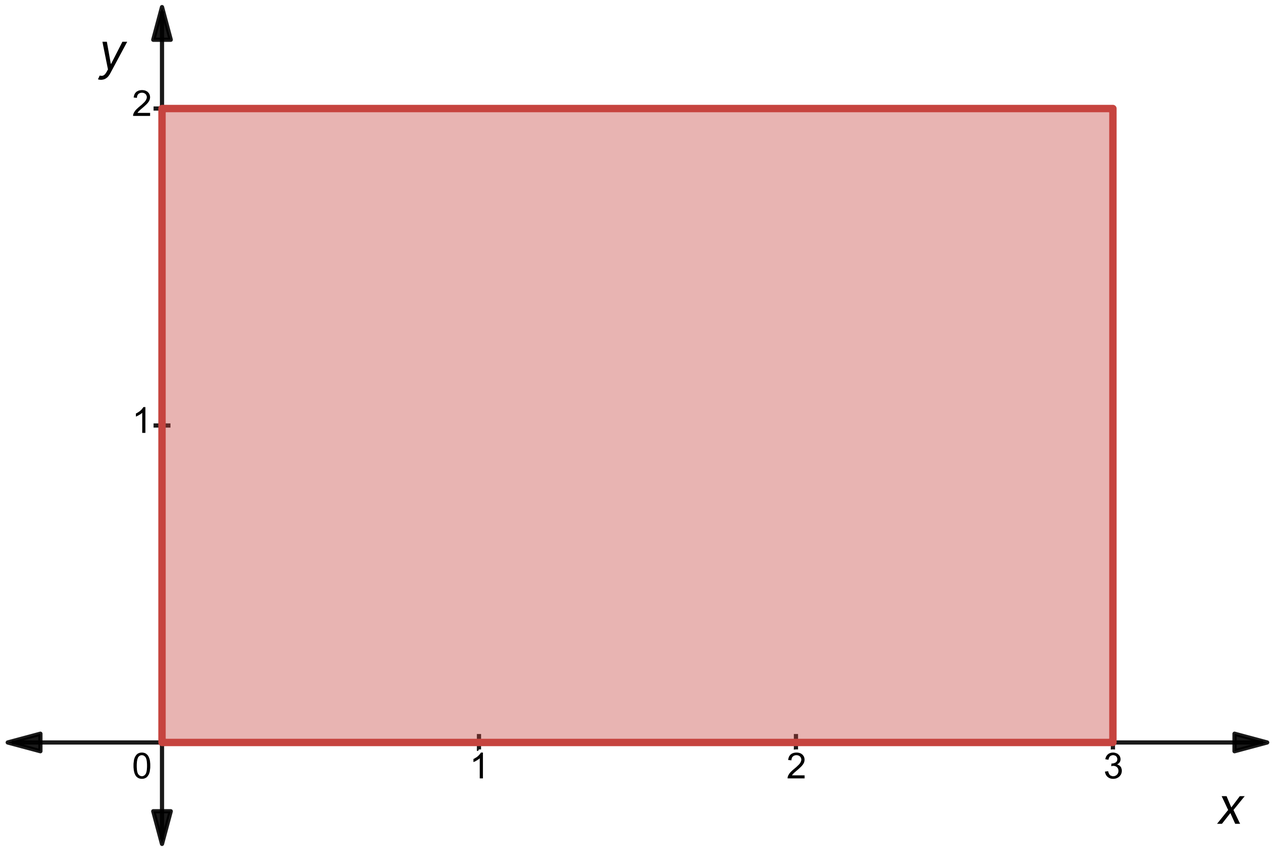
EECS · COSC594

5 May 2026

The Problem

Given a polygon as a set of points, rotate it along the y-axis and get the volume of the resulting object.

The Problem



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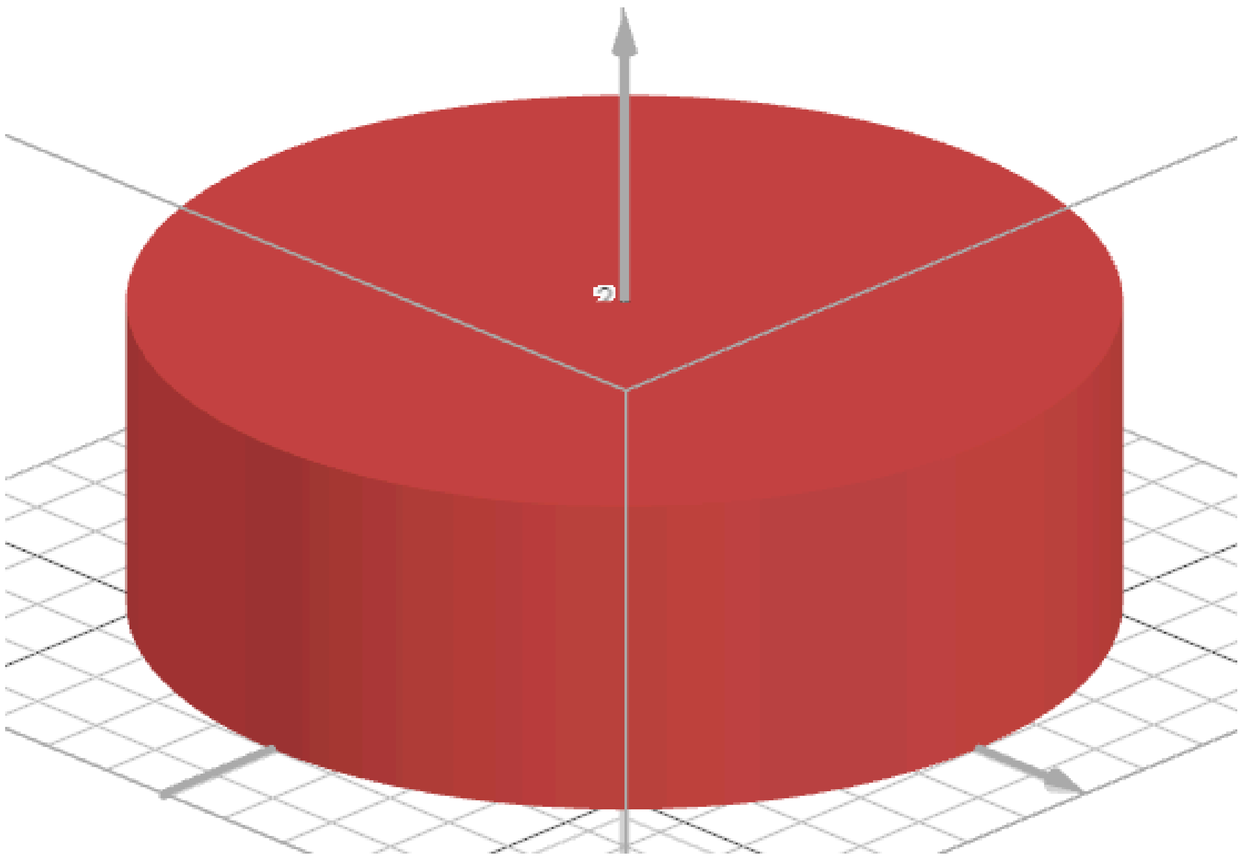


If we were to rotate around the y-axis, we get a cylinder with a radius of 3 and a height of 2.

$$V = \pi r^2 h = \pi(3^2)(2) = 56.55$$

Prototype and Constraints

Class: `PolygonRotation`
Method: `getVolume`
Parameters:

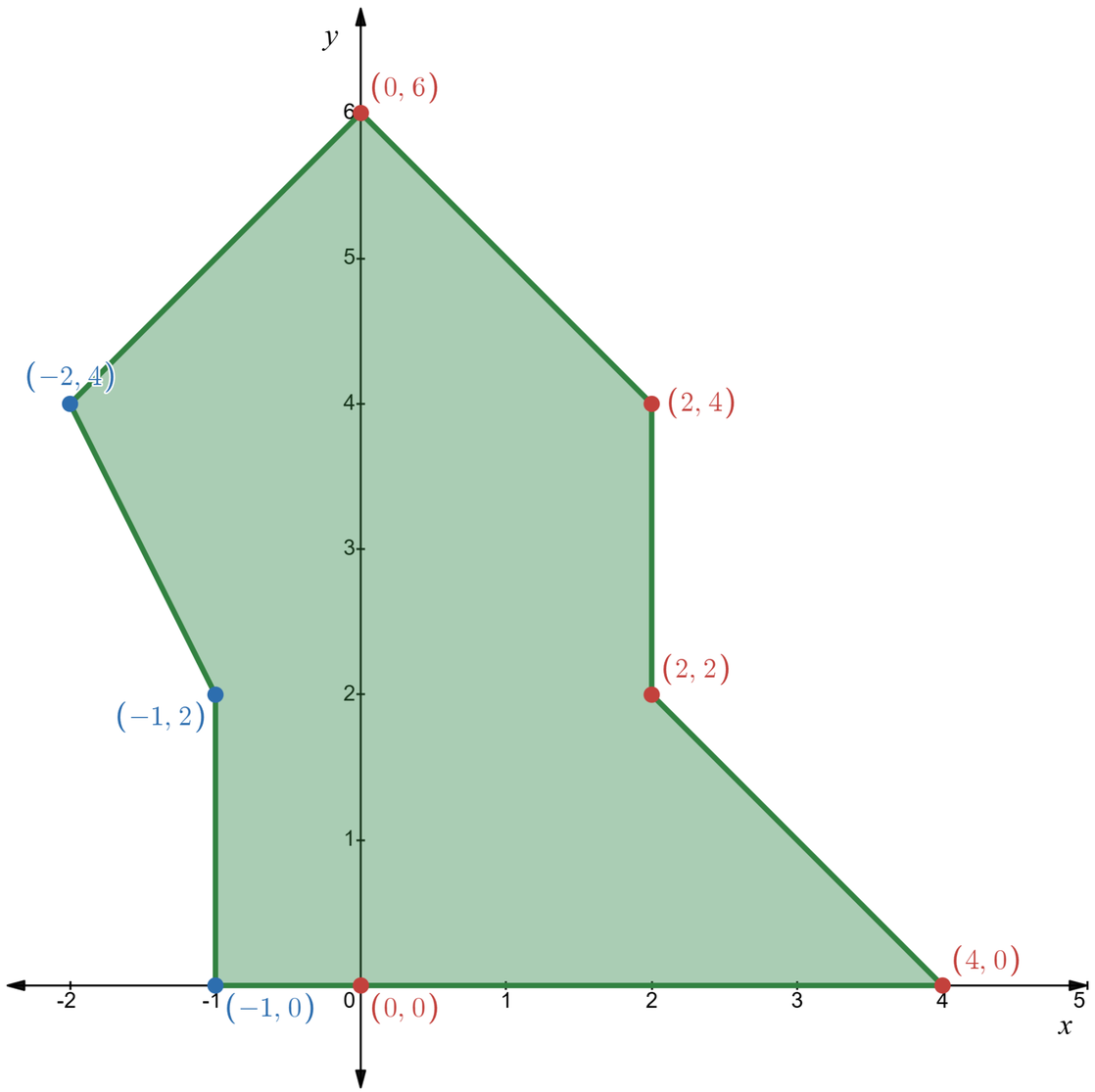
`int[]`	x
`int[]`	y

Return: `double`

- x will contain [3, 50] elements
- y will contain the same number of elements as x
- Each element of x and y will be [-100, 100]
- The polygon will be convex
- All vertices of the polygon will be distinct
- Only two points on the y-axis
  - (0, ymin) and (0, ymax) where ymin<ymax
- Vertices are listed in clockwise order starting with (0, ymax)
- y coordinates will be [ymin, ymax]
- No 3 points will be on the same line
- Returned value be within 1e-9 of the correct value

An Irregular Polygon

x	0, 2, 2, 4, 0, -1, -1, -2
y	6, 4, 2, 0, 0, 0, 2, 4



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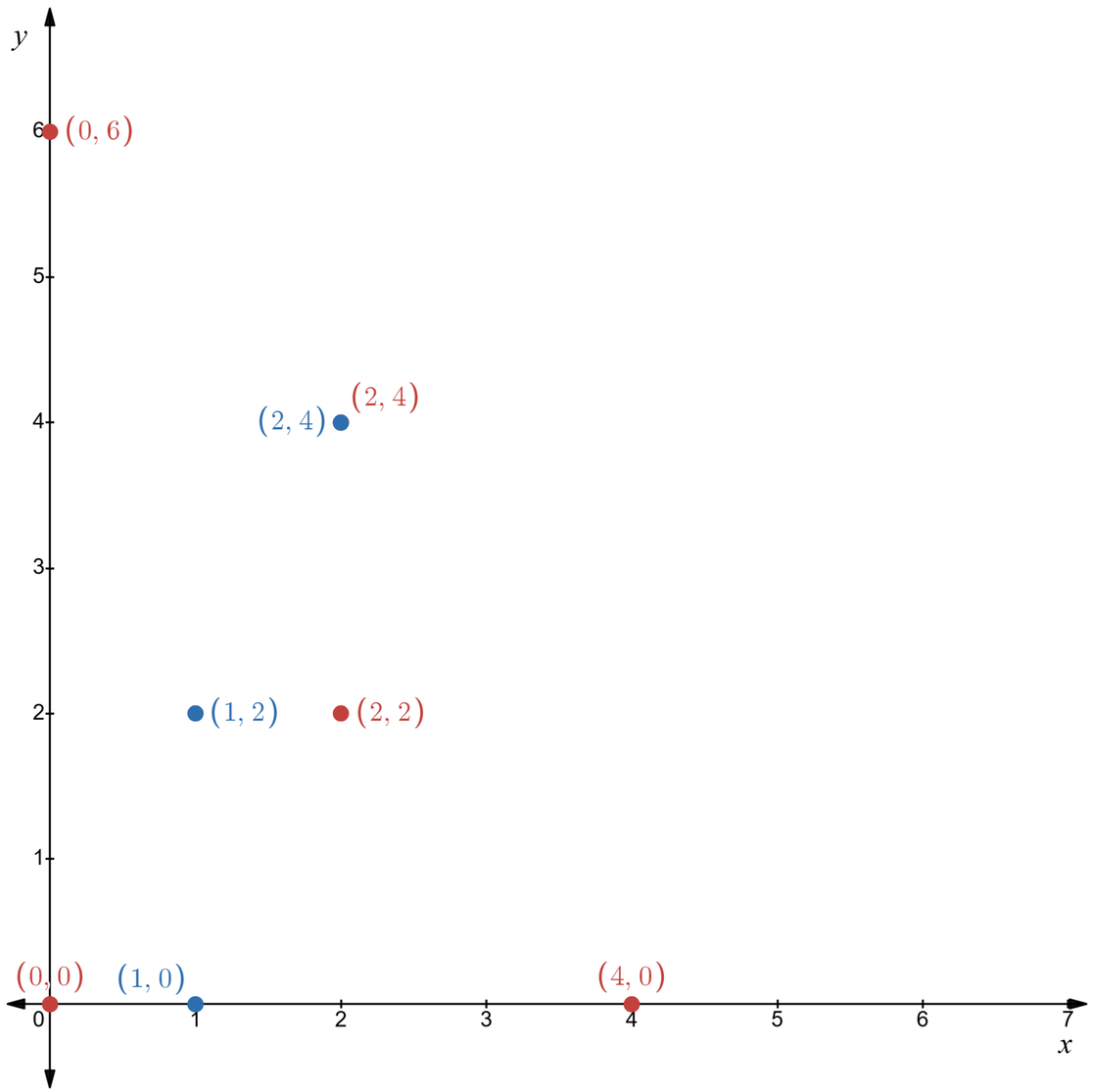
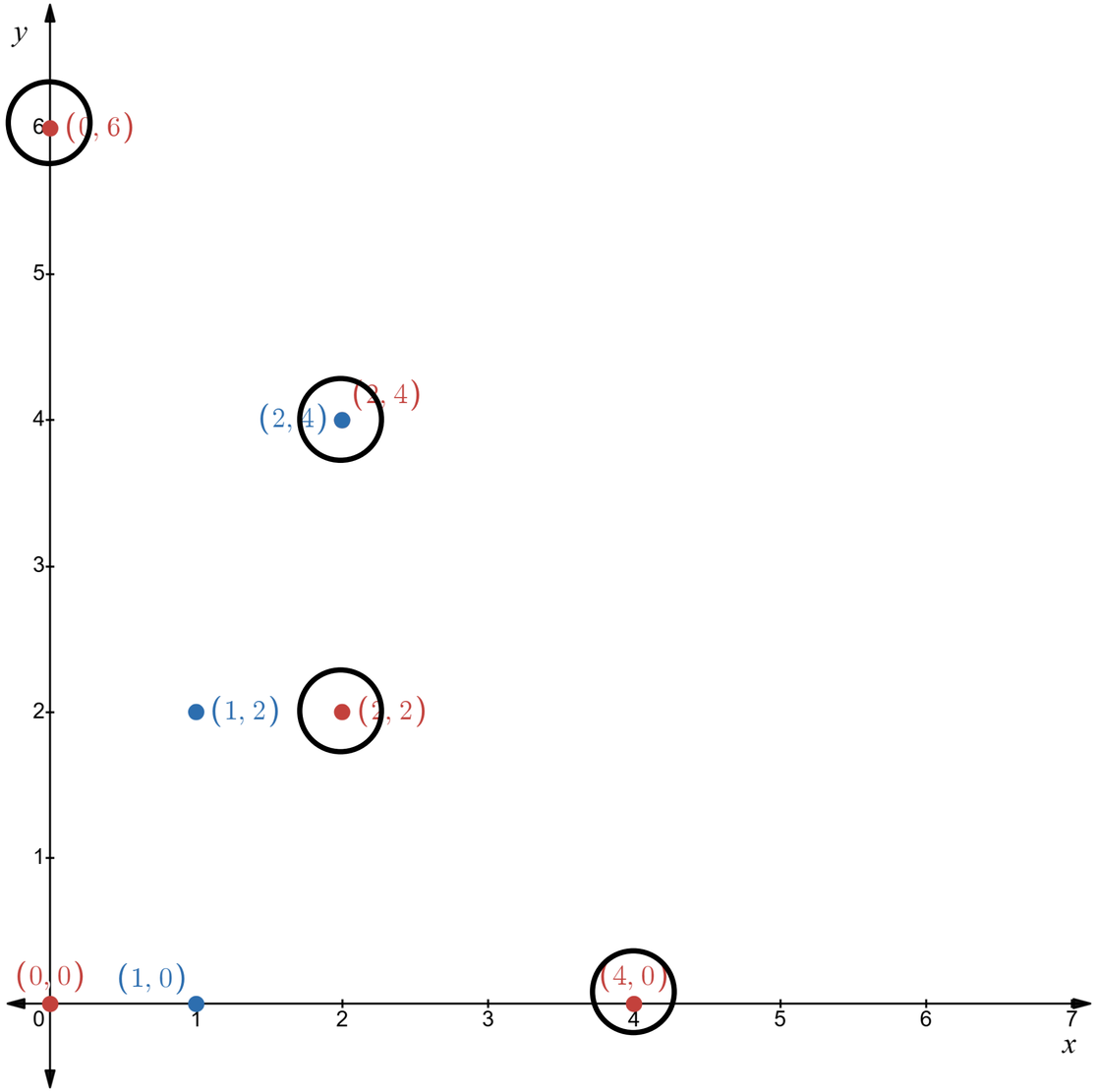
An Irregular Polygon



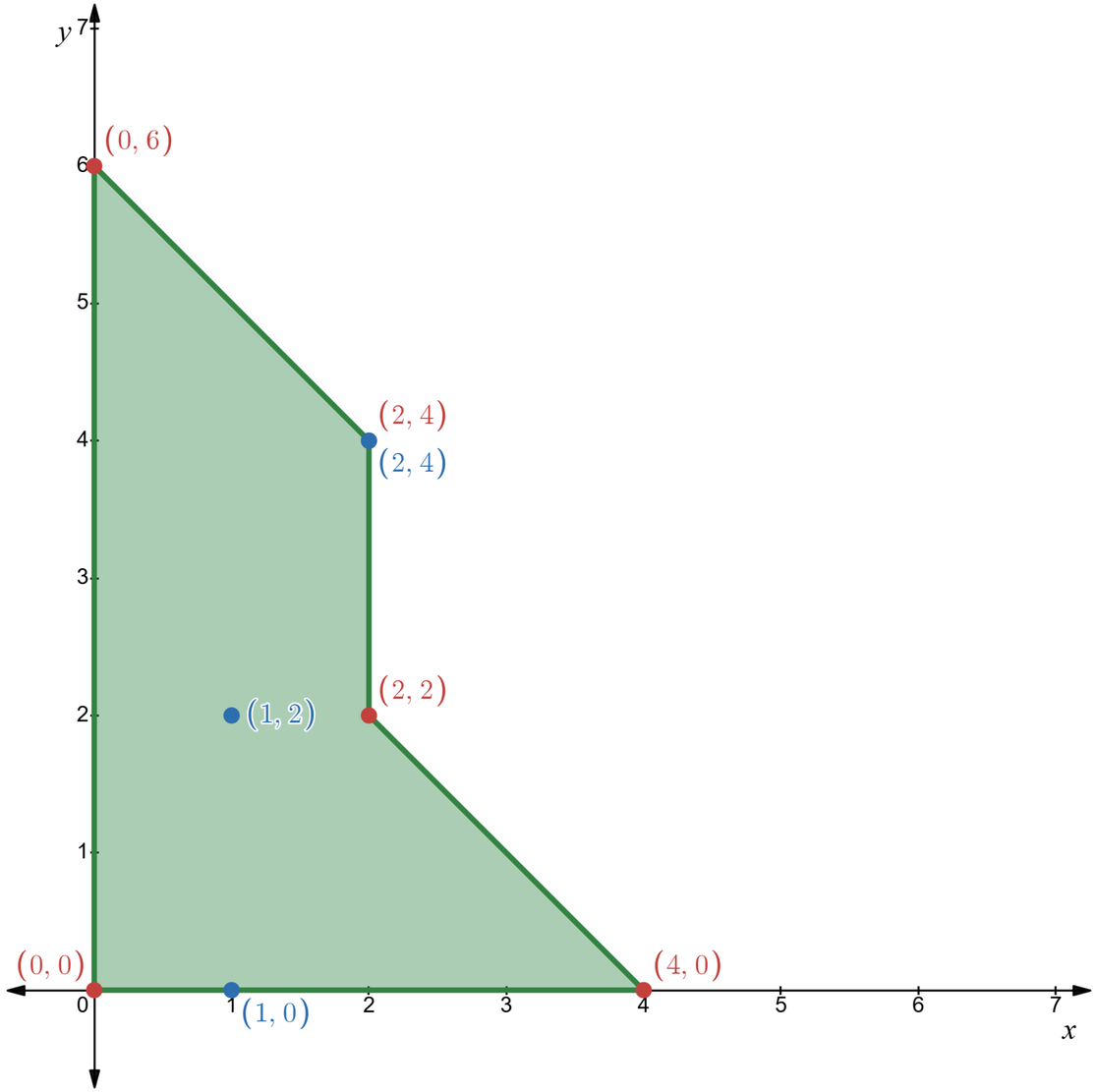
Make the negative points positive by taking their absolute value.

An Irregular Polygon



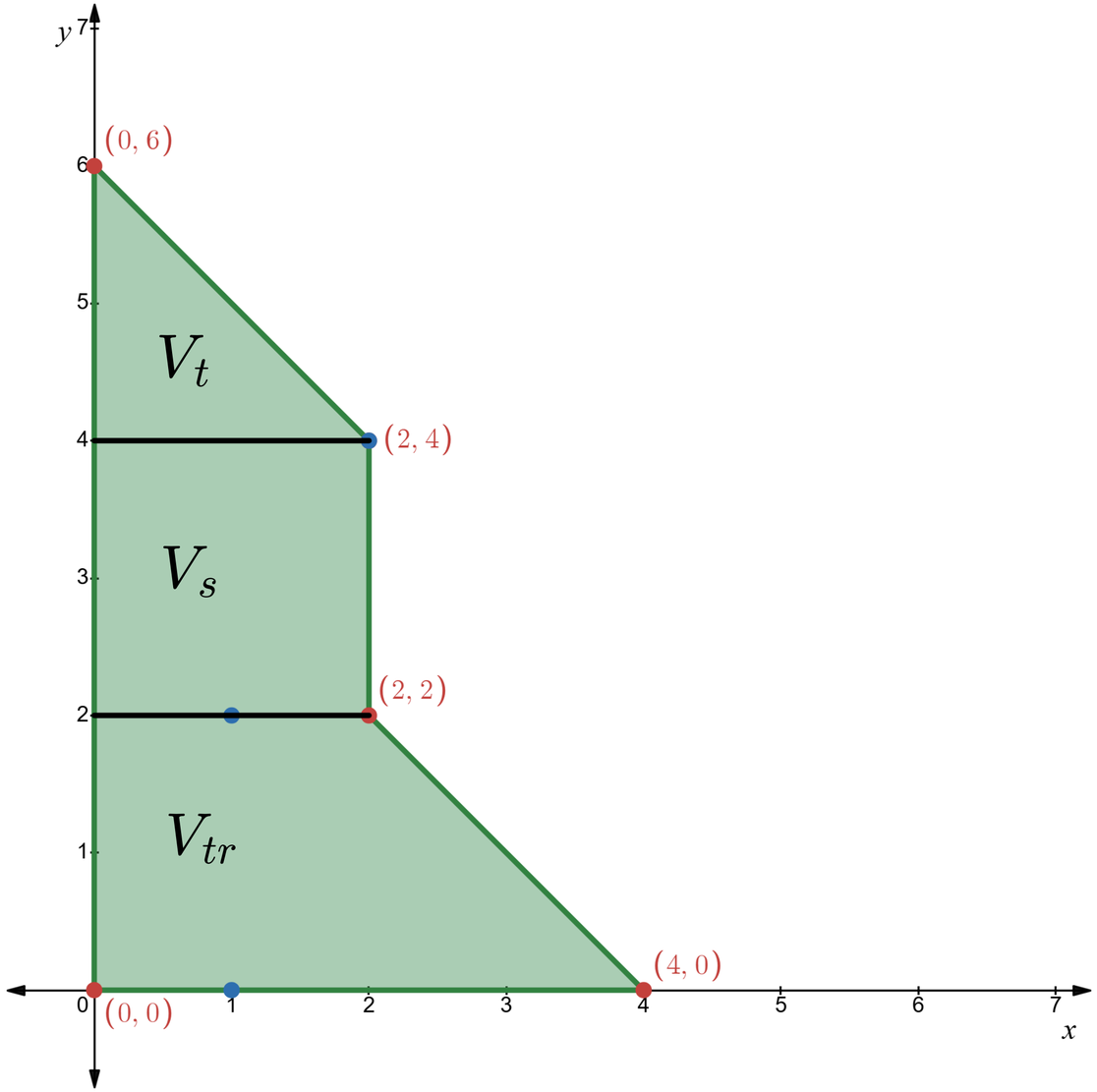
Now choose the outer most points of each y value.

An Irregular Polygon



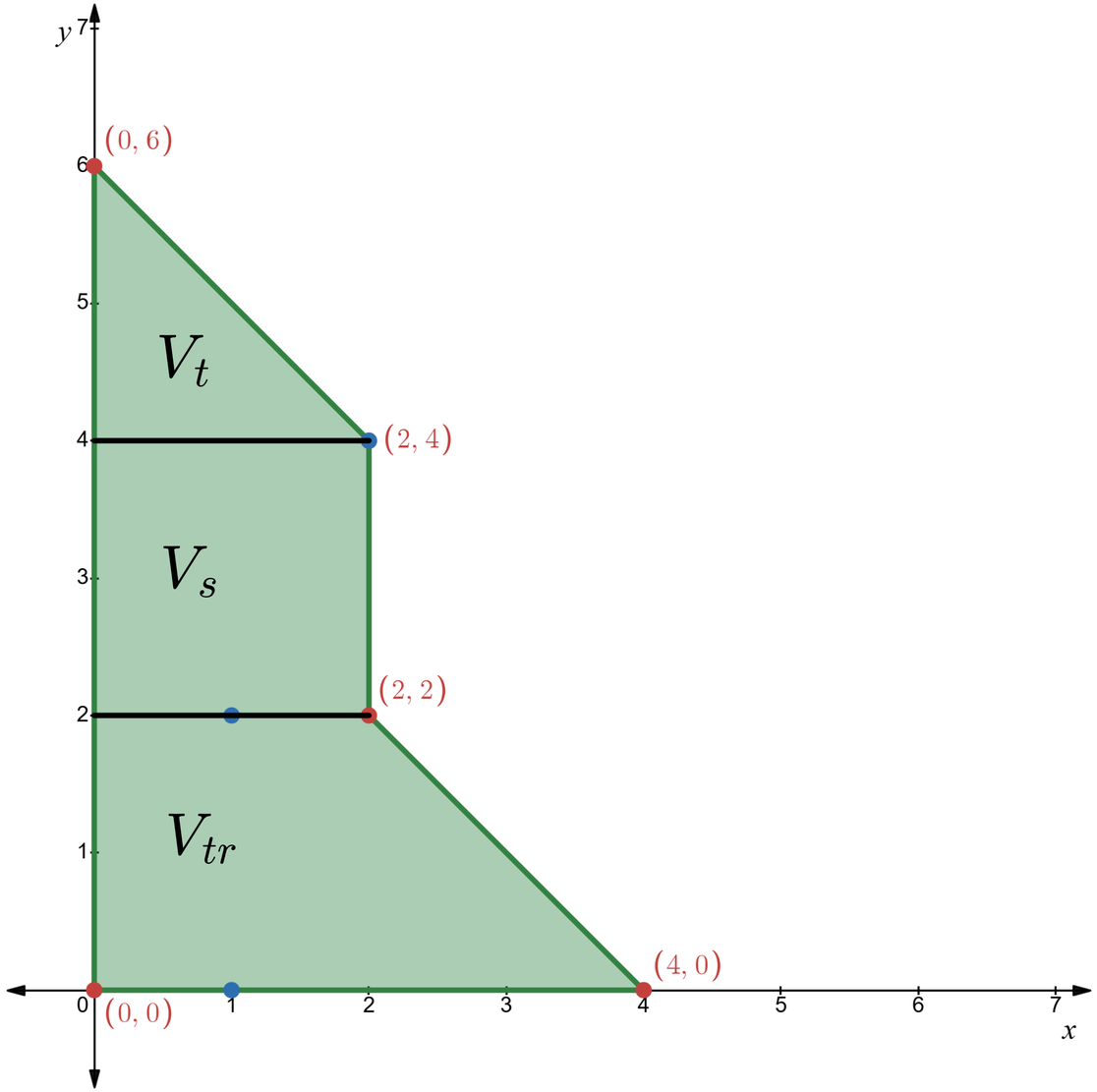
Now we have a new polygon that we can get the volume of.

An Irregular Polygon



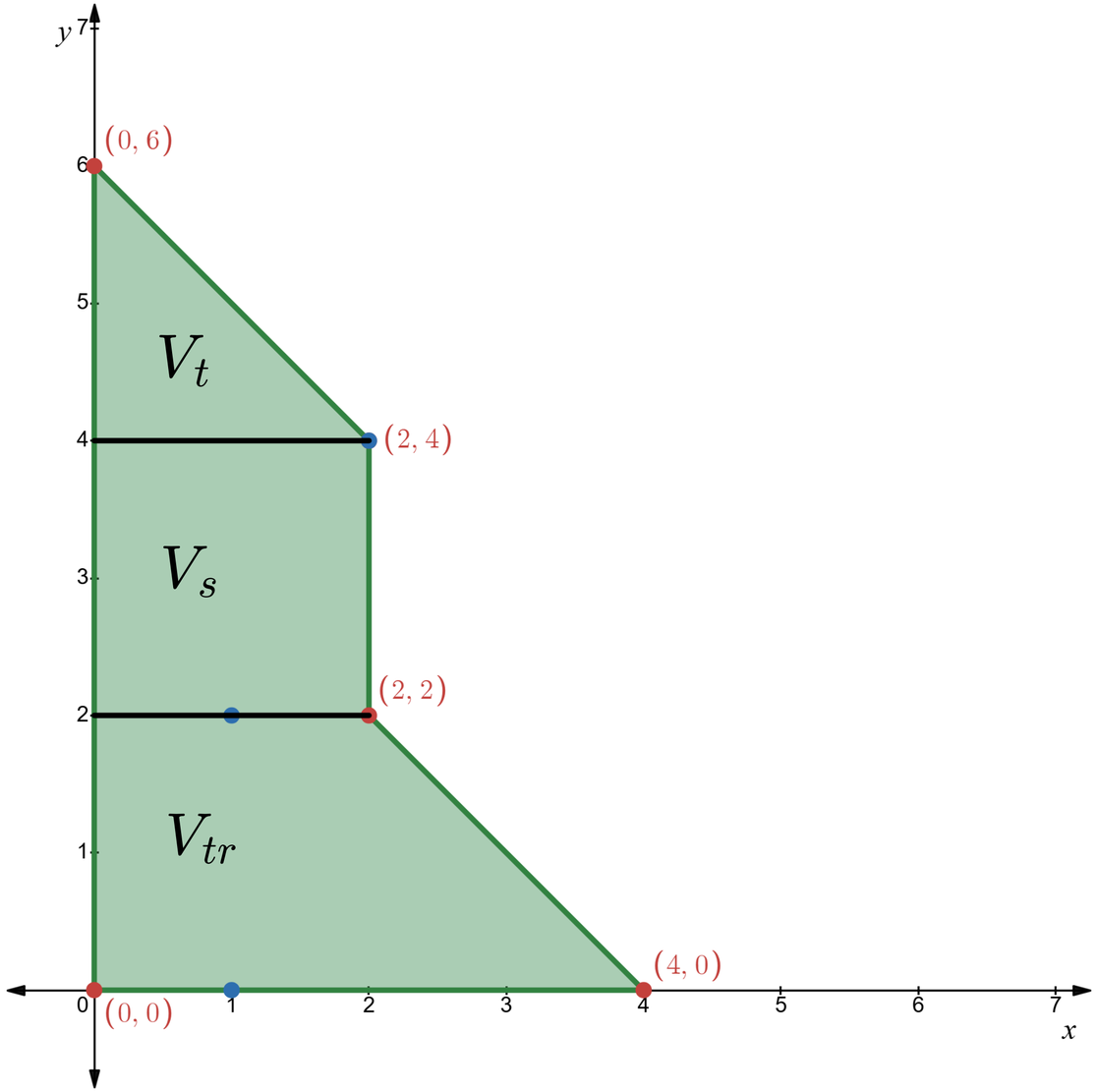
If we split the polygon along the y values, notice we have three simple polygons we can get the volume of.

An Irregular Polygon



If we split the polygon along the y values, notice we have three simple polygons we can get the volume of.
These are the only shapes that we can encounter: a triangle, a square, and a trapezoid.

An Irregular Polygon



However, we do not need to use three separate equations. We can use the frustum equation which accommodates all of these shapes.

$$V = \frac{1}{3}\pi h(r_1^2 + r_1 r_2 + r_2^2)$$

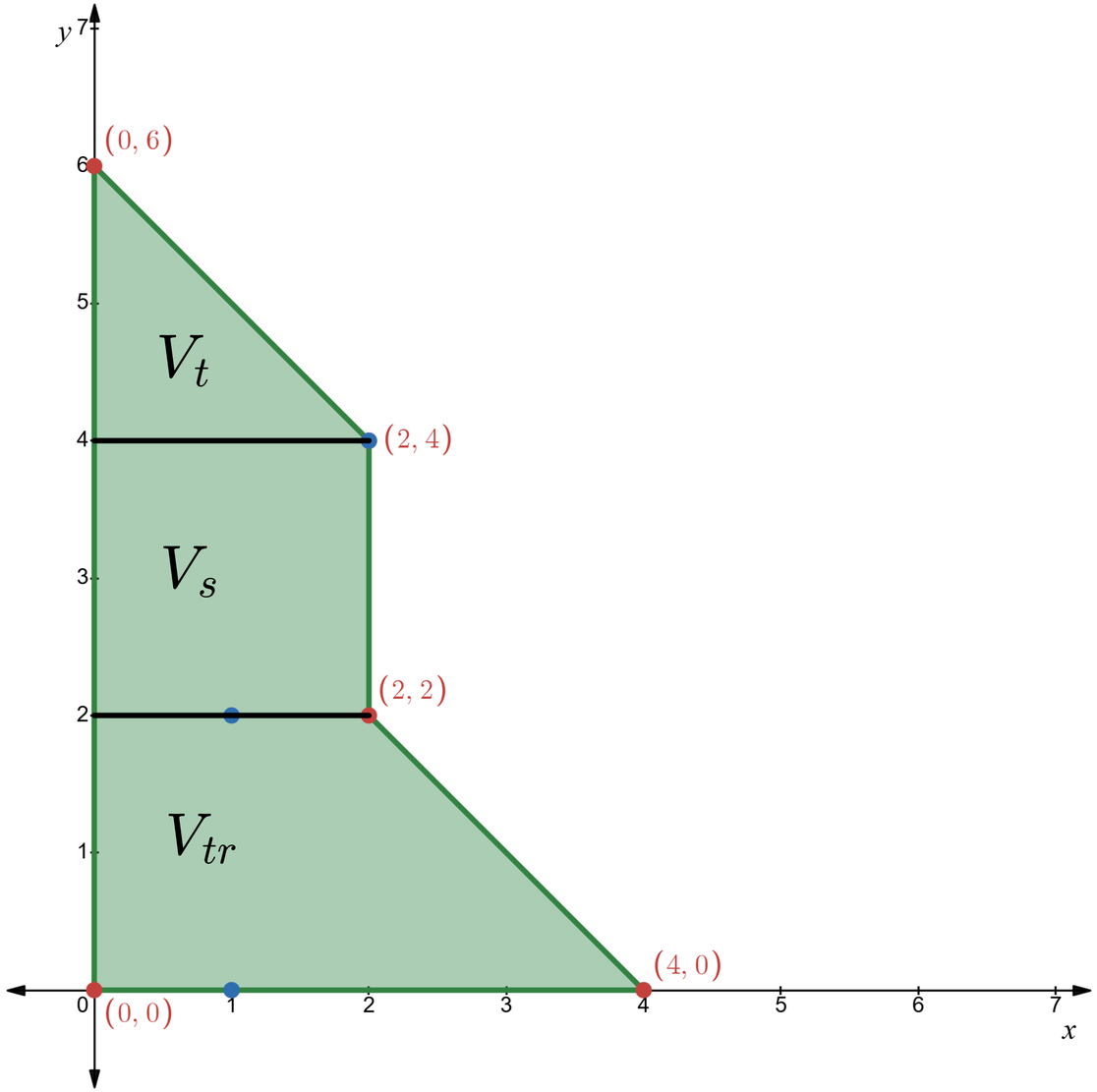
An Irregular Polygon

$$V_t = \frac{1}{3}\pi(2)(0^2 + (0)(2) + 2^2) = 8.3776$$

$$V_s = \frac{1}{3}\pi(2)(2^2 + (2)(2) + 2^2) = 25.1327$$

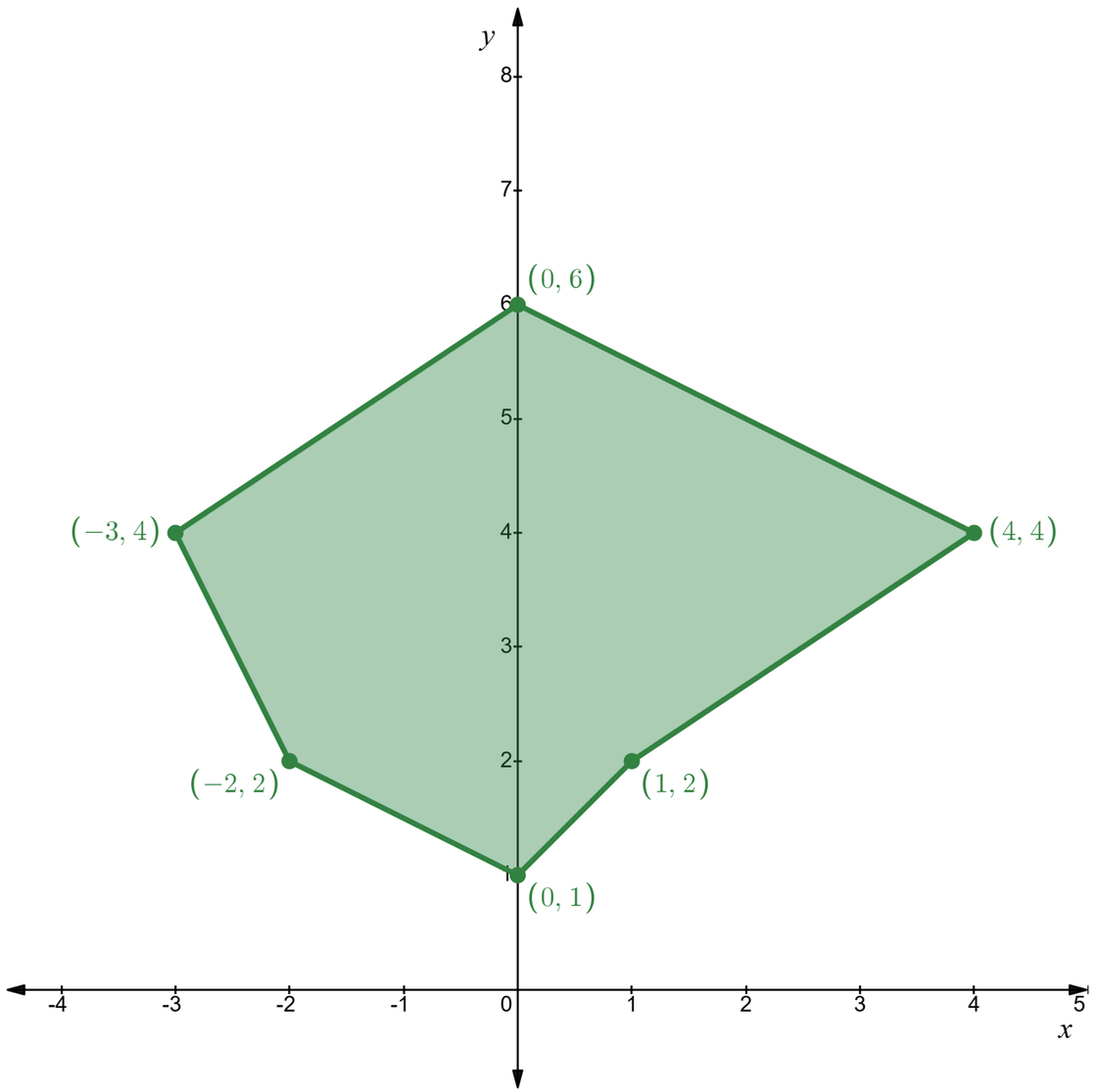
$$V_{tr} = \frac{1}{3}\pi(2)(2^2 + (2)(4) + 4^2) = 58.6431$$

$$V = 92.1534$$



Crossing Points

x	0, 4, 1, 0, -2, -3
y	6, 4, 2, 1, 2, 4



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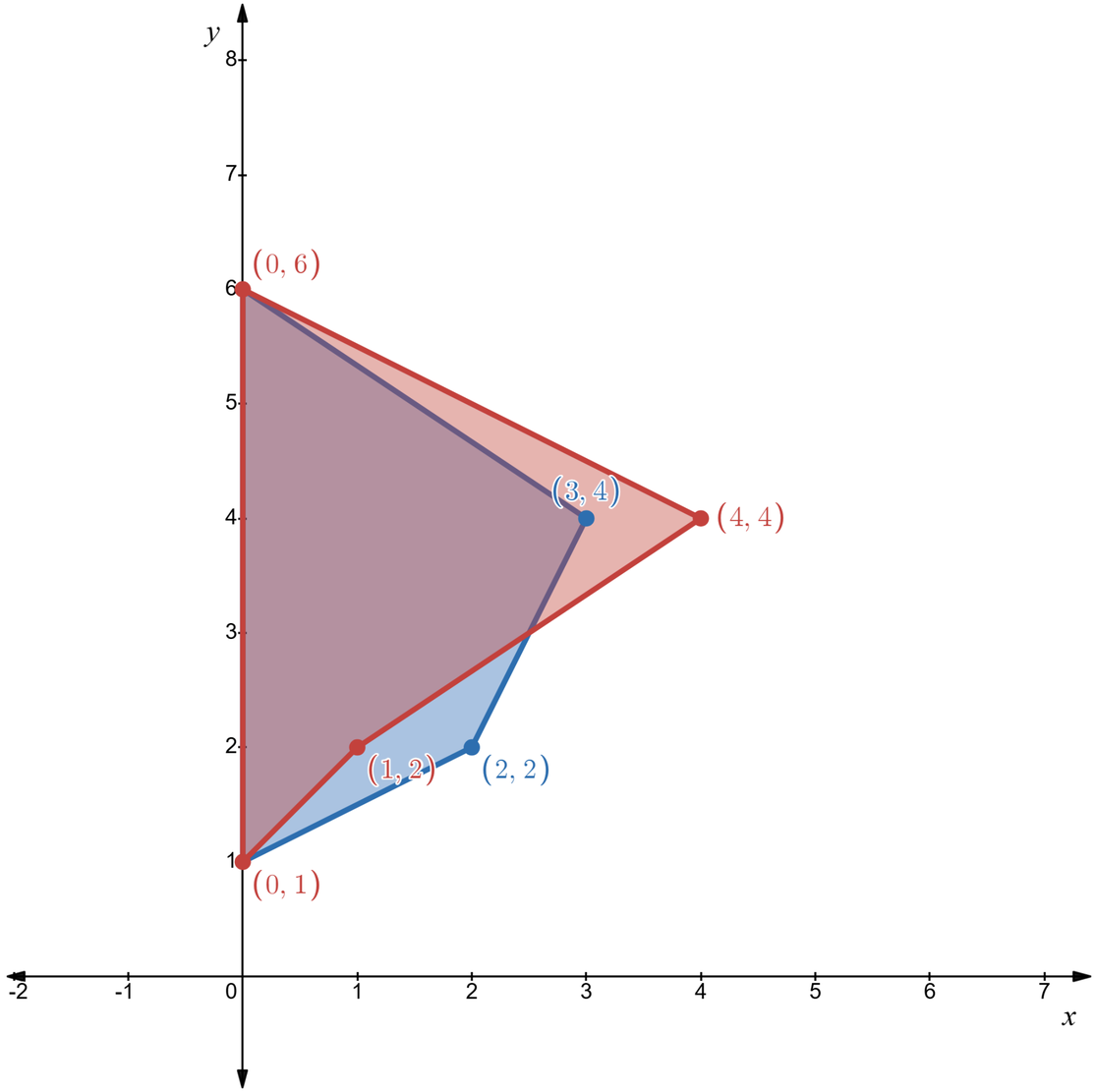
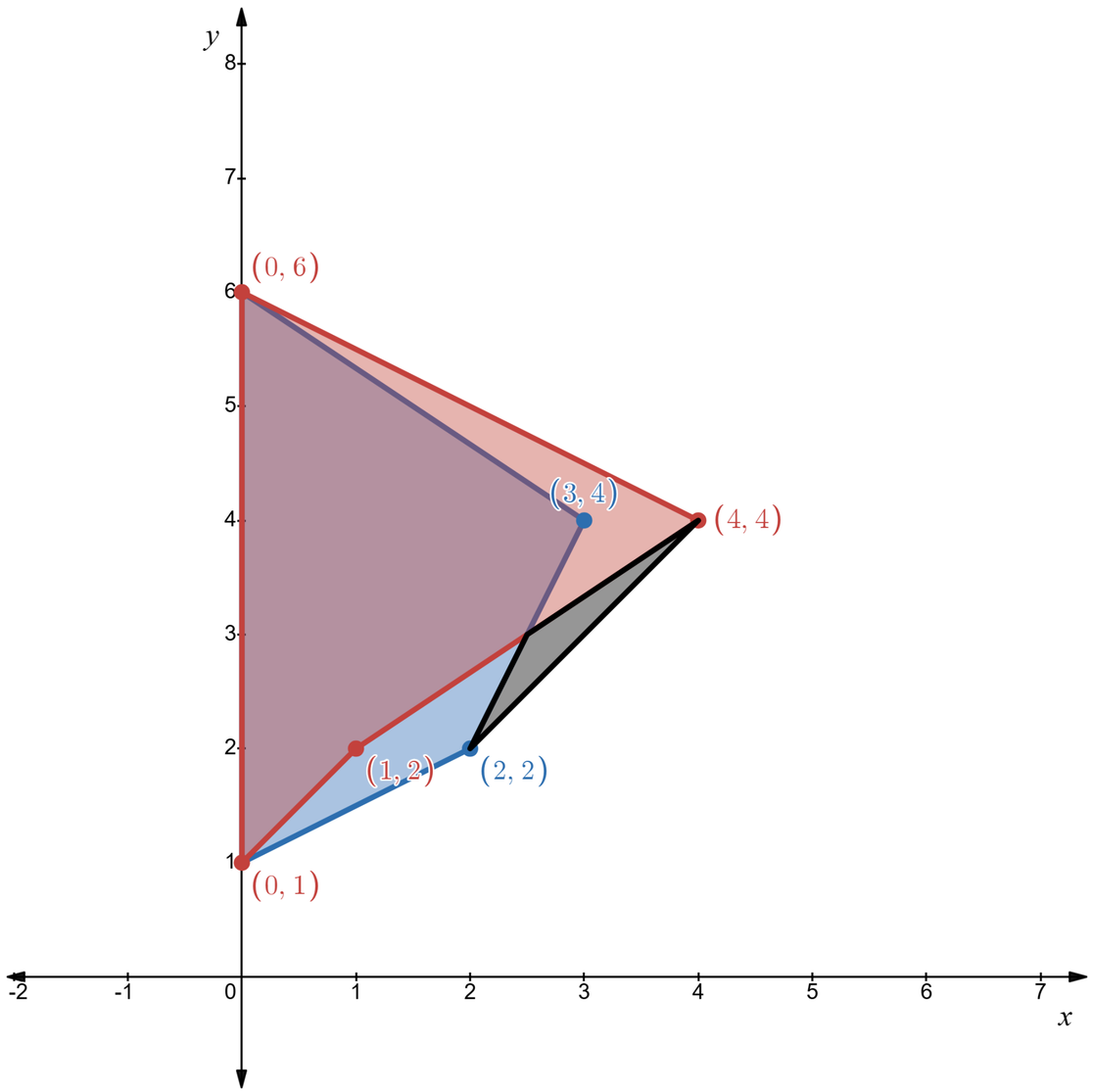
Crossing Points



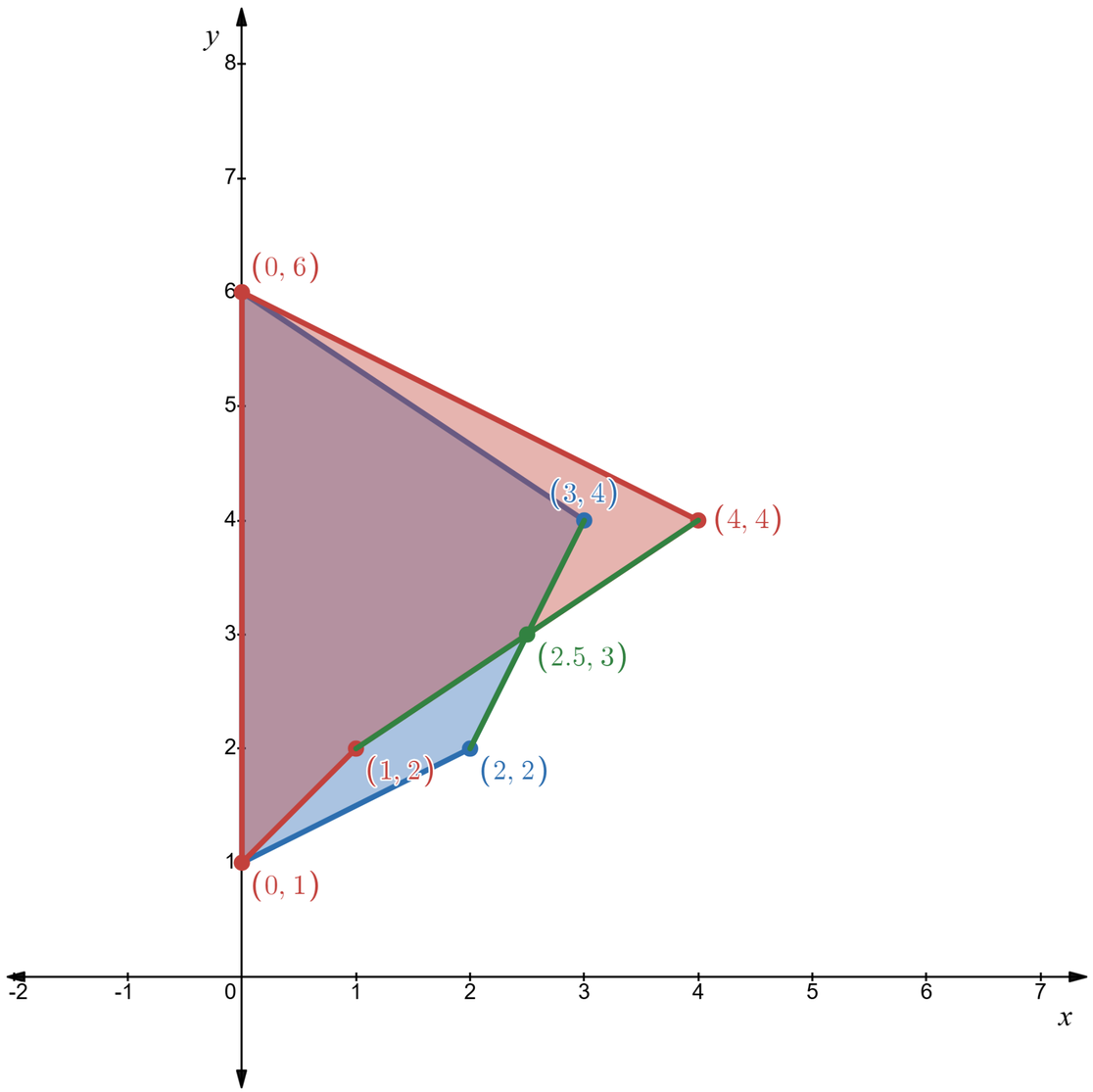
This time let's visualize the negative and positive polygons separately.

Crossing Points



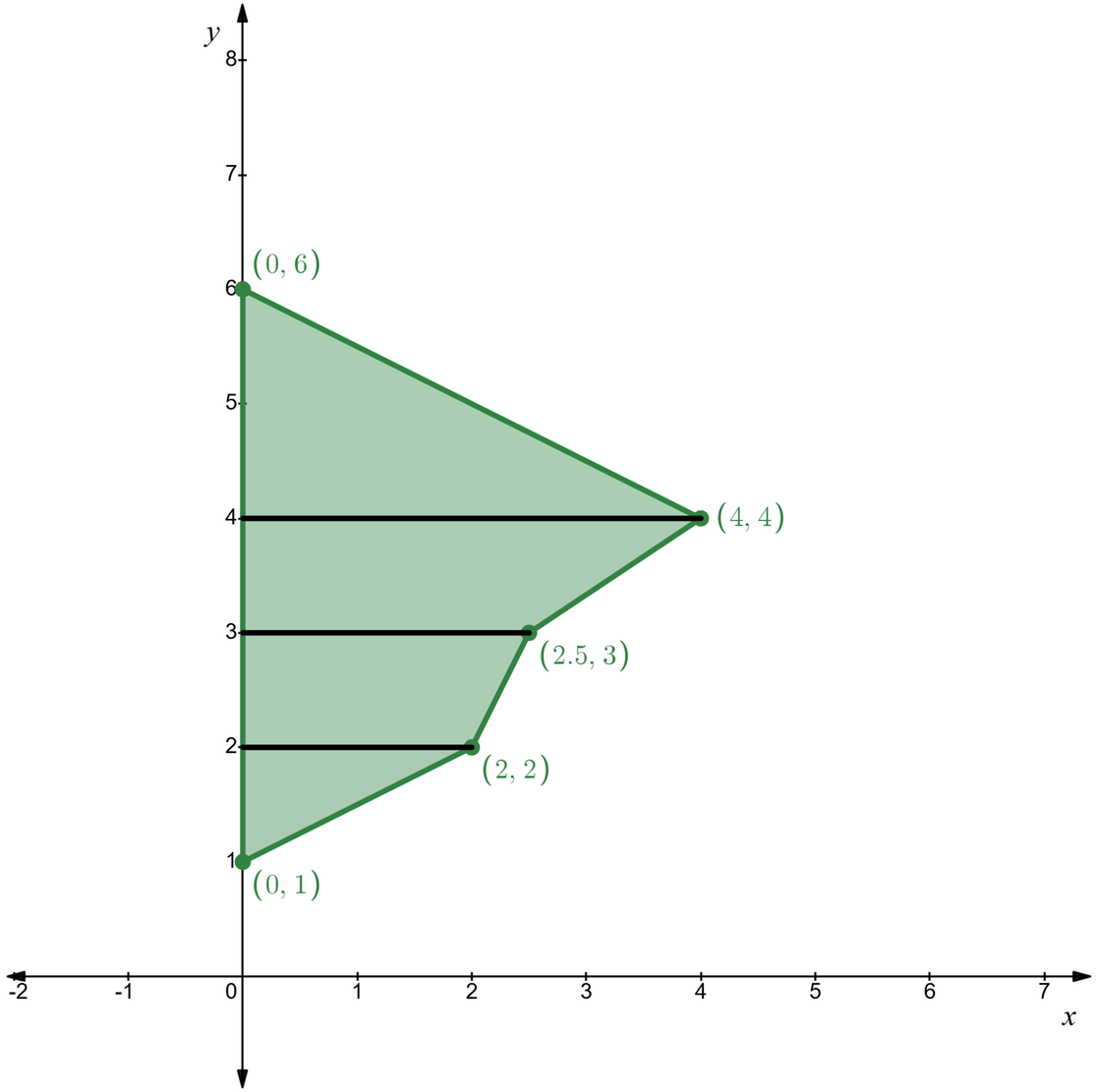
If we were to take the outer most points, we would add space that isn't apart of the object. So, we need to do some interpolation to find the crossing point.

Crossing Points



To find the crossing points, we check all combinations of line segments. If we find one, we add it to the list of points.

Crossing Points



Now we do the same as the last problem, take the outer most points at each y value. And calculate the volume by taking slices of the polygon.

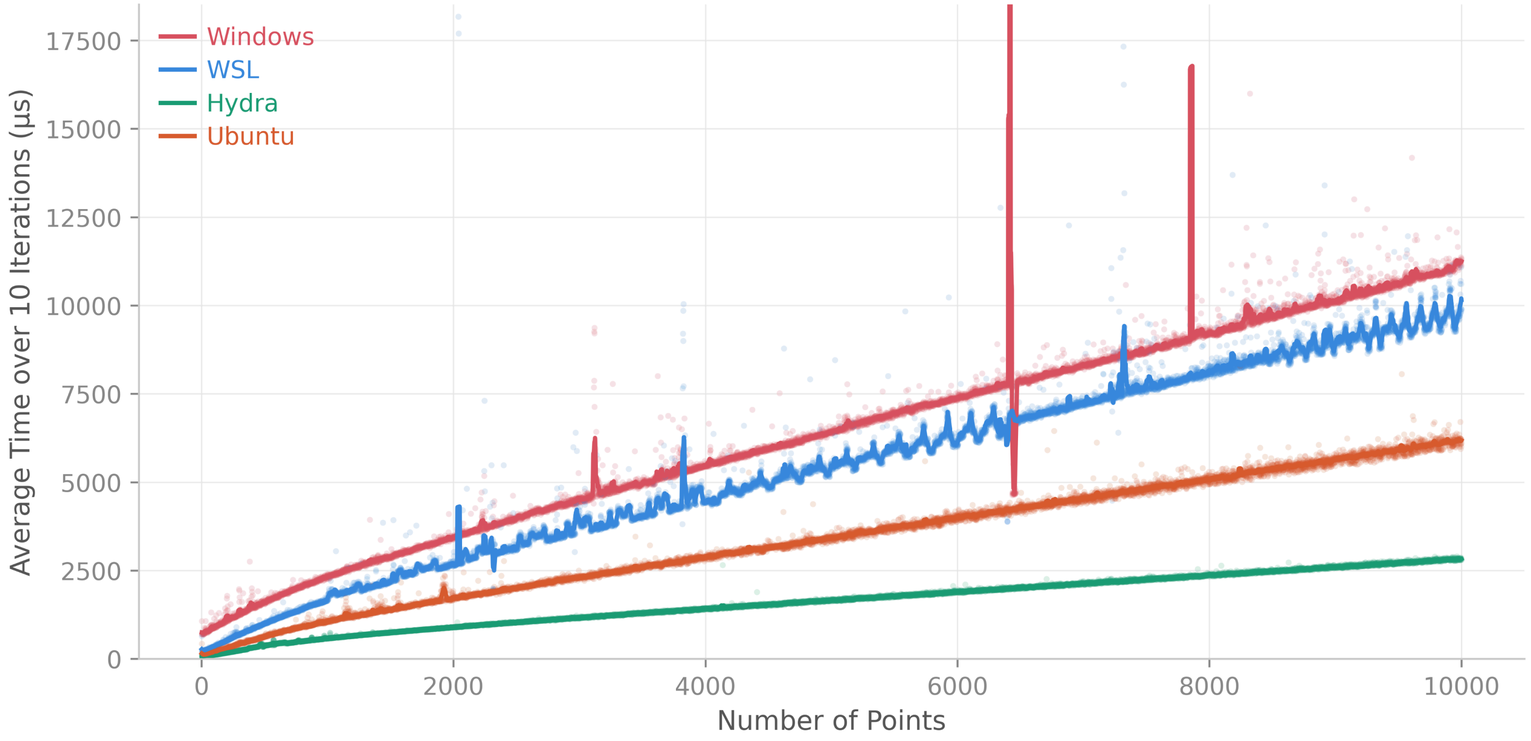
$$V = 87.441$$

Running Time

Separating the positive and negative points: $O(n)$
Checking for crossing points: $O(n \log n + n)$
Selecting the outer most points: $O(n)$
Calculating volume: $O(n)$

Overall: $O(n \log n)$

Performance



System Specs

- Windows, Ubuntu (WSL and Dual Boot) - AMD Ryzen 9 5900HX 3.30 GHz
- Hydra 0 - Intel Core i7-10700 4.8GHz

TopCoder Statistics

583
Opened the Problem

56
Submitted a Solution

19 **3.3%**
Correct Solutions

37:32 **20:21**
Average Correct Time

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